Feiyang **Chen**

AI Application & Systems Software Engineer

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Summary

University of Sydney computer science master's graduate with approximately three years of Linux kernel and systems software experience. Independently developed and maintain a news proofreading agent for in-house reporters and built an AI platform for model management, application integration, and usage tracking. Skilled in AI applications, OS porting, cross-architecture adaptation, and software debugging.

Skills

Languages	Python, C, Java
AI Applications	LangChain/LangGraph, RAG, MCP
Backend	FastAPI, Spring Boot, PostgreSQL, Redis
Engineering	Linux development, cross-architecture porting, Docker

Projects

AI News Proofreading and AI Platform

Suzhou, Jiangsu, China

Project Collaborator | Suzhou Xinsu Baoye Network Technology Co., Ltd. — *Apr. 2026 – Oct. 2026*

- **AI Proofreading:** Independently built a news proofreading agent with FastAPI and LangChain/LangGraph, using tool calls for fact retrieval and language/factual-consistency checks, with structured validation for applying suggested edits; used by in-house reporters, averaging 100+ calls per day.
- **Fact Retrieval:** Built news ingestion, entity/appointment extraction, and PostgreSQL retrieval, using alias matching and evidence update times to provide traceable evidence for time-sensitive news checks.
- **AI Platform:** Expanded the proofreading service into an AI platform, separating management from AI execution with Spring Boot and FastAPI; built an admin console and user workspace with model/Prompt configuration, application permissions, quotas, and usage reporting for proofreading and invoice recognition.
- **Business Systems and Integration:** Contributed to content management frontend/backend development, implementing multi-tenancy, legacy data migration, and third-party synchronization; integrated AI proofreading through the editor SDK and connected WorkBuddy via MCP/OAuth.
- **Model Deployment and Observability:** Deployed Qwen locally with vLLM and llama.cpp and integrated it into the platform; traced model and tool calls with OpenTelemetry and cached proofreading results by configuration and fact versions.
- **Personal Technical Validation:** Validated LangGraph routing/query rewriting and BM25/vector retrieval with RRF in a separate open-source A-share analysis agent; evaluated faithfulness and tool use with DeepEval and documented architecture choices and lessons learned.

Systems Software and Open-source Contributions

Suzhou, Jiangsu, China

Kernel and Systems Development (Including Open-source Maintenance) — *Jun. 2020 – Feb. 2025*

- Maintained a separate LoongArch64/Linux branch of Valgrind; independently implemented libVEX support for the base instruction set and adapted the runtime, and contributed to vector instruction support; reached initial usability within one month, passed 100+ regression tests, and added platform/nightly CI test configurations, advancing low-level toolchain compatibility with LoongArch.
- Contributed to Linux system call and network driver porting; added LoongArch32/64 support to nolibc, statx compatibility, and platform selftests, merged into mainline Linux; added LoongArch support to Kcov, merged upstream.
- In separately maintained branches, added LoongArch/QEMU support to RT-Thread, contributed to LS2K0300 integration, and ported Newlib.

Work Experience

Loongson Technology Corporation Limited

Suzhou, Jiangsu, China

Kernel Engineer — *Dec. 2020 – Aug. 2023*

- Contributed to MIPS/LoongArch Linux kernel development and upstream patch submissions, most merged into mainline; diagnosed defects, implemented fixes, and ran regression tests.
- Led LoongArch porting for Valgrind and other foundational software, resolving cross-architecture build, instruction compatibility, runtime, and performance issues.
- Implemented CI/CD for the company's documentation repository, automating document builds and preview website generation so community members could preview changes and collaborate on documentation.

Jiangsu Lemote Technology Co., Ltd.

Suzhou, Jiangsu, China

Kernel Engineer — *Jun. 2020 – Dec. 2020*

- Developed and debugged Linux kernel code through issue reproduction, log analysis, patching, and validation.
- Ported OpenStack Nova and DevStack to Fedora MIPS in separately maintained branches, resolving component compatibility and environment issues; brought up the core service workflow within one month.

Education

The University of Sydney

Sydney, NSW, Australia

Master of Computer Science — *Feb. 2024 - May 2026*

- **ISSRE 2026 accepted paper (second author):** "Dr. DD: 1-Minimal Isolation of Failure Causes via Deferred Restarts."
- **Implementation and Experiments:** Built a Python delta debugging framework to automatically reduce failure-inducing inputs for binary software, aiding bug reproduction and diagnosis; validated it on Binutils/Valgrind, comparing reduction quality, test calls, and execution time.

Suzhou University of Technology (formerly Changshu Institute of Technology)

Suzhou, Jiangsu, China

Bachelor of Computer Science and Technology (Industry-integrated Program) — *Sep. 2016 - Jun. 2020*

- **ICAIS 2020 paper (first author):** "Formal Verification of the Correctness of Operating System on Assembly Layer in Isabelle/HOL."
- **Research:** Modelled MINIX 3's i386 context-saving assembly in Isabelle/HOL and verified register/memory states and instruction execution against expected results.